

Resolution Geometry on Chaotic and Dirty Experiments

Declared Regularity, Certified Window Atlases, Anchor Budgets,
and Ergodic Transfer of the Metre

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Abstract

The theorem ladder [1] conditions its differential and metric layers on regularity (H-reg), on a metrological anchor (H-SI) and on a resolved medium (H-med). Two objections follow: real data are not regular, and the anchor hypotheses move the engineering problem to the user. This paper tests both objections on chaotic and contaminated experiments. Two parts of the critique are confirmed. First, the noiseless asymptotic law of a chaotic map is discontinuous in its parameter, and short deterministic records are mutually singular, so the differential layers do not apply to that experiment. Second, a counted channel removes dilation only if its loss mechanism is not scale-covariant; multiplicative count loss reopens the unit fibre.

The remaining parts are refuted. Regularity is a property of the declared experiment, not of the latent dynamics. Walls are located by windowed certification with controlled error. The calibration burden is computable: an anchor budget equals the dimension of invariant forms of the empirical isotropy. Medium nuisance is a kernel direction removed by frequency or path diversity, and integer ambiguities are a discrete fibre resolved by a ratio-tested search.

On a chaotic slice with heavy tails, outliers, dropouts, cycle slips, a turbulent dispersive medium, outages and three detector regimes, the procedure certifies 19 of 30 windows in metres with median error 0.49% (maximum 0.68%) and abstains elsewhere; a naive global calibration has median error 6.5%, and an all-or-nothing pipeline rejects the record. Across an outage, a stationary chaotic source carries the metre through an unknown dilation and anisotropy. A clock-based invariant flags source drift, and the residual error of unflagged drift is bounded by a measured sensitivity floor (1.0%). Eight registered checks pass on four seeds, and four mutation canaries are caught.

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1 The critique

C1 (regularity). Real data are rarely regular or dominated, so Sections 5–11 of the ladder do not apply, and a chaotic, dirty experiment will trip on a wall.

C2 (anchor burden). Trusting that a counted channel removes dilations (H-SI) and that the medium is resolved (H-med) shifts an unquantified burden to the user.

Claim	Verdict	Section
Noiseless chaotic experiments are irregular and non-dominated	confirmed	2
Therefore RG cannot be applied to chaotic systems	refuted	2
Dirty data yield no output	refuted	3, 8
Calibration effort is unquantified	refuted	4
H-med is an act of faith	refuted	5
A counted channel always removes dilation	confirmed (gap)	6
The unit is lost during an anchor outage	bounded	7

Table 1: Verdicts. “Bounded” means crossed under a testable hypothesis with a measured error floor.

2 Regularity is declared

No-Go 2.1 (Noiseless chaos). For the logistic family $x \mapsto rx(1-x)$, the law map $r \mapsto \lim_T \frac{1}{T} \sum_t \delta_{x_t}$ is not continuous: across $\Delta r = 10^{-3}$ at the period-three window ($3.828 \rightarrow 3.829$), the total-variation distance between long-run histograms is 0.568 and the Lyapunov exponent changes from 0.301 to -0.171 . Hyperbolic parameters are dense [4, 9], and linear response fails for this family [2]. For $x_0 \sim U(0, 1)$ the laws of (x_0, x_1) lie on the disjoint curves $x_1 = rx_0(1 - x_0)$, so distinct parameters give mutually singular laws. The experiment is neither regular nor dominated.

The exact layer of the ladder (L1–L11) needs no regularity and applies unchanged: the quotient is the set of distinct laws, and here every parameter is its own class.

Imported Theorem 2.2 (Regularity of noisy finite-horizon experiments; [14, 6]). *Let the declared record be $y_{0:T}$ with $y_t = m_t(\theta) + \varepsilon_t$, where $T < \infty$, the maps $\theta \mapsto m_t(\theta)$ are C^1 , and ε_t are independent with a smooth positive density of finite Fisher information. Then the experiment is dominated and differentiable in quadratic mean. The same holds for Markov records with Gaussian dynamical noise of fixed variance.*

Proposition 2.3 (Declared noise regularizes; MC02). *With dynamical noise $\sigma = 10^{-2}$, the total-variation distance between the two parameters of No-Go 2.1 is 0.011, and the per-step Fisher information $\mathbb{E}[(x(1-x))^2]/\sigma^2$ varies over $r \in [3.80, 3.86]$ with maximal relative jump 0.94%. It*

scales as σ^{-2} (4, 400, 4×10^4 for $\sigma = 10^{-1}, 10^{-2}, 10^{-3}$), which locates the non-dominated wall at $\sigma = 0$.

Remark 2.4. The geometry obtained is that of the declared experiment. It depends on σ and T , as the ladder requires (L32). What is refuted is the inference from “the latent dynamics are chaotic” to “the experiment is irregular”.

3 Walls are located, not fatal

Definition 3.1 (Certified window atlas). Partition the record into windows. A window is *certified regular* if a lower confidence bound of the smallest singular value of the estimated response exceeds a declared tolerance τ . The certified windows, with walls between them, form the atlas on which the ladder’s differential layers are applied.

Proposition 3.2 (Controlled certification; MC03). *Consider a three-channel, two-parameter response observed through Student- t_3 noise with 5% gross outliers. Use Huber regression [5] and bootstrap lower bounds [3] at level $\alpha = 0.05$ with $\tau = 0.3$. In 100 wall windows and 100 regular windows, the false-certification rate was 0.00 and the power 1.00. At 60% outliers the certification fails, as it must.*

Lemma 3.3 (Invariance is certified by equivalence, not by non-rejection). *A statement “structure X is unchanged” may be certified only by an equivalence test: an upper confidence bound of the change must lie below a declared tolerance [11]. Failure to reject a change is not certification.*

Remark 3.4. In the slice of Section 8, replacing the equivalence test of detector isotropy by non-rejection lets a stale calibration be carried into an anisotropic regime, with distance-ratio errors above 2% (canary MC07). With 200 samples per window the equivalence test certifies no carry-forward: the data cannot establish unchanged isotropy to 2%, and the procedure abstains.

4 The anchor budget

Theorem 4.1 (Anchor budget; MC04). *Let h be a constant metric on a target chart V whose empirical isotropy group is $K \leq GL(V)$. Radar anchors $\Delta_a^\top h \Delta_a = \rho_a^2$ determine h iff their number in general position is at least*

$$b(K) = \dim(\text{Sym}^2 V^*)^K.$$

With fewer anchors, a family of dimension $b(K) - k$ fits them.

Proof. The admissible metrics form the linear space $(\text{Sym}^2 V^*)^K$. Each anchor is one linear equation in h ; in general position k equations have rank $\min(k, b(K))$. $\square$

Corollary 4.2. *In two dimensions $b = 1$ for $SO(2)$ and for C_4 , and $b = 3$ for C_2 or for no symmetry. In three dimensions $b = 1$ for an irreducible rotation group and $b = 6$ for none. Computed ranks with $k = 1, \dots, 4$ anchors are 1, 1, 1, 1 ($SO(2)$) and 1, 2, 3, 3 (C_2). With two anchors and no symmetry, $h = I$ and $h = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}$ fit both anchors and differ by 0.6 on a held-out pair.*

The number of anchors a dirty experiment needs is therefore a computed output of its empirical symmetry, not an open-ended effort.

5 The medium as a kernel direction

Proposition 5.1 (Frequency diversity; MC05). *For two-way delay $\tau(f) = 2d/c + \sum_{p \in P} 2K_p/f^p$, the parameters (d, K_p) are identifiable iff the number of frequencies is at least $|P| + 1$, with the Vandermonde-type rank condition. With $P = \{2\}$, one frequency gives rank 1, and two give rank 2 (condition number 9.2 for 1.5 and 1.2 GHz). With $P = \{2, 3\}$, two frequencies leave an exact kernel direction, and three give rank 3 (condition number 848). An unmodelled dispersion term is an unremoved fibre.*

Proposition 5.2 (Path diversity; MC05). *For non-dispersive excess path with range $\rho_e = \rho_0 - h \sin e + Z/\sin e$ over elevations e [10], a single elevation leaves (h, Z) confounded. Over twelve elevations from $e_{\min}$ to 90° the condition number is 2322, 110, 37.7, 28.7 for $e_{\min} = 60^\circ, 30^\circ, 15^\circ, 5^\circ$. The cost of H-med is this number.*

Definition 5.3 (Integer fibre). After loss of lock, a counted carrier record carries an unknown integer offset per channel. The integer vector is a discrete nuisance fibre of the joint law.

Proposition 5.4 (Ratio-tested resolution; MC07; [12, 15]). *Nonlinear integer least squares on the accumulated trilateration misclosure, accepted only when the ratio of second-best to best cost exceeds 3, resolved every re-acquisition in the slice of Section 8 (three stations: 3 accepted, 0 wrong; four stations: 3 accepted, 0 wrong). Until acceptance the procedure abstains. Redundancy raises the ratio margin: 21.5 with four stations versus 6.0 with three at the first re-acquisition.*

6 When a counted channel does not remove dilation

No-Go 6.1 (Scale-covariant count loss; MC06). Let the counted channel record $n = kd$ with $k = 2\Delta\nu_{Cs}/c$ and loss mechanism L . If L is multiplicative, $n' = (1 - \epsilon)kd$ with unknown ϵ , then (d, ϵ) has an exact dilation fibre: Fisher rank 1 in (scale, ϵ) over any set of anchor distances. Additive slips $n' = kd - s$ and dead time $m = n/(1 + n\tau_d)$ [7] give rank 2 when the anchor distances, and hence the count rates, vary.

Hypothesis 6.2 (H-SI, refined). In addition to the clauses of the ladder, the count-loss mechanism of the anchor channel is either absent or not scale-covariant, and its parameters are identified jointly with the anchor.

This clause was missing from the ladder. Without it, Lemma L8 (count rigidity) is true, but its consequence L48 (dilation removed) can fail.

7 Crossing an outage with a chaotic source

During an anchor outage, a change of detector scale is a pure dilation and is invisible to every imaging-only natural module (L47). One further structure can carry the unit: a stationary source.

Hypothesis 7.1 (H-stat). Time translation belongs to Γ on the source channel across the outage: the source law, and hence its invariant measure μ , is the same before and after.

Theorem 7.2 (Ergodic transfer). *Assume H-stat, and that the calibrated segment gives physical positions with covariance Σ and third central moment tensor T of μ . Let $G_T = \{L : L\Sigma L^\top = \Sigma, L \cdot T = T\}$. If the outage record is $y = Mp + b + \text{noise}$, then M is determined by the second and third moments of y up to right multiplication by G_T . If $G_T \subset O(2)$, all distances $\|M^{-1}\Delta y\|$ are determined.*

Proof. $\text{Cov}(y) = M\Sigma M^\top$ gives $M = \text{Cov}(y)^{1/2}O\Sigma^{-1/2}$ with O orthogonal. The third moments give $M \cdot T = T_y$; two solutions differ by $L \in G_T$. If L is Euclidean-orthogonal, $\|L^{-1}p\| = \|p\|$. $\square$

Remark 7.3. The condition $G_T \subset O(2)$ is checkable on the calibrated segment. For the Lorenz attractor the residual symmetry is the reflection $x \mapsto -x$, an isometry.

Proposition 7.4 (Stationarity test). *Two families of invariants are unaffected by any spatial linear map and test H-stat across the outage: (i) affine-invariant standardized moments of the whitened record; (ii) the mean interval between maxima of the whitened radius, read on the clock channel, which runs during the outage. Family (ii) is the powerful one. At record length 3000 time units, a drift $\rho : 28 \rightarrow 30$ gives $z = -13.2$ on (ii) and 1.8 on (i).*

Theorem 7.5 (Transfer certificate; MC08). *A transfer is certified if neither invariant family is flagged ($|z| \leq 3$). The certificate carries an error floor ε^* : the largest distance error among unflagged drifts at the given record length. At 3000 time units, with an unknown dilation of 3% and anisotropy 1.25:*

ρ after outage	median distance error	period z	moment $\max z $	flagged
28.00	0.50%	-0.6	0.9	no
28.15	0.60%	-1.6	1.2	no
28.50	1.01%	-2.9	0.4	no
29.00	1.96%	-7.1	0.6	yes
30.00	3.76%	-13.2	1.8	yes

so $\varepsilon^* = 1.0\%$, and no drift with error at least 3% went unflagged. On seeds 1, 42 and 777, ε^* was 0.57%, 0.46% and 0.47%, and $\rho = 28.5$ was flagged. Without the clock invariant, the drift to $\rho = 30$ (error 3.8%) is not flagged (canary MC08).

Remark 7.6 (What is crossed and what is not). The unit is carried across the outage only under H-stat, which is tested but not proved by the invariants. Drift below their sensitivity biases the result by at most ε^* . That bound is a boundary of the data, not of the method, and it shrinks with record length. Without a stationary source, the outage dilation is not crossed.

8 A chaotic dirty slice

Construction (MC07).

- *Source.* A target follows the Lorenz attractor [8] projected to a 21×19 m region.
- *Radar.* Three stations (then four) range the target by two-way dual-frequency timing (1.5 and 1.2 GHz, 30 ps noise) through a turbulent dispersive medium with about 3 m of delay. Cycle slips of ± 1 –3 cycles occur at a rate of 1/400, and there are three radar outages.
- *Imaging.* Student- t_3 position jitter (2 cm scale), 3% gross outliers and 2% dropouts.
- *Detector regimes.* 6000 samples in three regimes: isotropic; anisotropic ($R(0.3) \text{diag}(1, 1.25)$); anisotropic with a 3% pixel-pitch dilation.

Procedure per window (200 samples).

1. Repair cycle slips continuously within each tracking segment.
2. After re-acquisition, resolve the integer fibre with the ratio test; abstain until it is resolved.
3. Form the ionosphere-free combination and trilaterate.
4. Certify the radar by its misclosure.
5. Calibrate the detector robustly on the first half-window; certify by held-out distance prediction on the second half (1%).

6. In outages, carry the calibration only under an equivalence test of detector isotropy (Tyler scatter [13]) (Lemma 3.3); otherwise abstain.

	3 stations	4 stations
windows certified in metres	19/30	19/30
median / max distance error, certified	0.49% / 0.68%	0.45% / 0.98%
outage windows certified in metres	0	0
wrong integer fixes accepted	0	0
naive global calibration: median / 90th percentile	6.5% / 18.2%	9.8% / 23.6%
all-or-nothing certification	reject	reject

Table 2: Window pattern (three stations): MMMMMMMM...M...MM...MMMM.MM, where M is certified in metres and “.” is abstention. Abstentions are outages and windows awaiting integer resolution. Seeds 1, 42, 777: 13–17 windows certified, maximum certified error $\leq 0.91\%$, 0 wrong fixes.

The procedure does not certify the record as a whole; it certifies an atlas. Where it abstains, the reason is named: an outage, an unresolved integer fibre, or unestablished isotropy.

9 Procedure for chaotic and dirty experiments

1. *Declare the experiment.* Include the noise model and the horizon; regularity is a property of this declaration (Imported Theorem 2.2).
2. *Window the record.* Certify regular windows with lower confidence bounds; walls are abstentions (Proposition 3.2).
3. *Test Γ per window.* Certify invariance only by equivalence tests (Lemma 3.3).
4. *Compute the anchor budget $b(K)$* from the empirical isotropy (Theorem 4.1).
5. *Remove medium fibres.* Use frequency diversity for dispersive terms and path diversity for non-dispersive ones; report condition numbers.
6. *Treat integer ambiguities as a discrete fibre.* Accept a fix only by ratio test.
7. *Check the count-loss mechanism* against refined H-SI (Hypothesis 6.2).
8. *Across outages,* transfer the unit only under H-stat, with clock-based and affine invariants, and report the floor ε^* (Theorem 7.5).
9. *Validate every certified window on held-out anchors.*

10 Boundaries that remain

1. **Noiseless chaos.** There is no differential geometry of the noiseless asymptotic chaotic experiment (No-Go 2.1).
2. **Scale-covariant count loss.** It reopens the unit fibre unless the loss is identified (No-Go 6.1).
3. **Outage dilation without a stationary source.** It is not crossed.
4. **Sub-threshold source drift.** It biases an ergodic transfer by up to ε^* .

5. **Unresolved integer fibres.** With poor geometry diversity, or with offsets beyond the search, the procedure abstains.
6. **Unmodelled dispersion, or non-dispersive excess with no path diversity.** These are exact kernel directions.

11 Ladder amendments

Where	Amendment	Status
after L8	count rigidity does not imply dilation removal under scale-covariant loss	no-go
H-SI	add the count-loss clause (Hypothesis 6.2)	hypothesis
after L51	anchor budget $b(K) = \dim(\text{Sym}^2 V^*)^K$	Γ -conditional
after L11	invariance certified only by equivalence tests	forced (statistical)
after L55	ergodic transfer under H-stat, with error floor	hypothesis-conditioned

A Reproducibility

`rg_chaos_attack.py`, with seed 20260930 and seeds 1, 42 and 777, passes 8/8 registered checks. `--canary` sabotages MC02 ($\sigma = 0$, infinite Fisher information), MC03 (60% outliers), MC07 (equivalence test disabled) and MC08 (clock invariant removed); each is caught.

Four statements were corrected during development and are recorded in the JSON:

1. Per-window slip repair left accumulated offsets; it was replaced by continuous tracking and integer resolution.
2. Linearized integer search produced near-unit ratios; it was replaced by nonlinear integer least squares.
3. The registered claim that three stations cannot resolve the integer fibre was an artifact of that search and was withdrawn.
4. The prediction that $\rho = 28.5$ is flagged failed at the primary seed ($z = -2.9$). The certificate was re-registered as an error-floor statement.

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